

Topic 2: Membranes, Proteins, DNA and Gene Expression

2.1 Properties of Gas Exchange Surfaces + Fick's Law

Essential Properties of Efficient Gas Exchange Surfaces:

1. Large surface area

- Increases space for diffusion
- Example: alveoli in lungs (≈300 million)

2. Thin barrier (short diffusion distance)

- Speeds up diffusion

- Example: alveolar epithelium is **one cell thick**

3. Steep concentration gradient

- Maintained by **ventilation and blood flow**
- Ensures continuous diffusion of gases

4. Moist surface

- Gases (e.g., O₂, CO₂) dissolve before diffusing
- Important for solubility and speed of diffusion

Fick's Law of Diffusion:

$$\text{Rate of diffusion} \propto \frac{\text{Surface Area} \times \text{Concentration Gradient}}{\text{Diffusion Distance}}$$

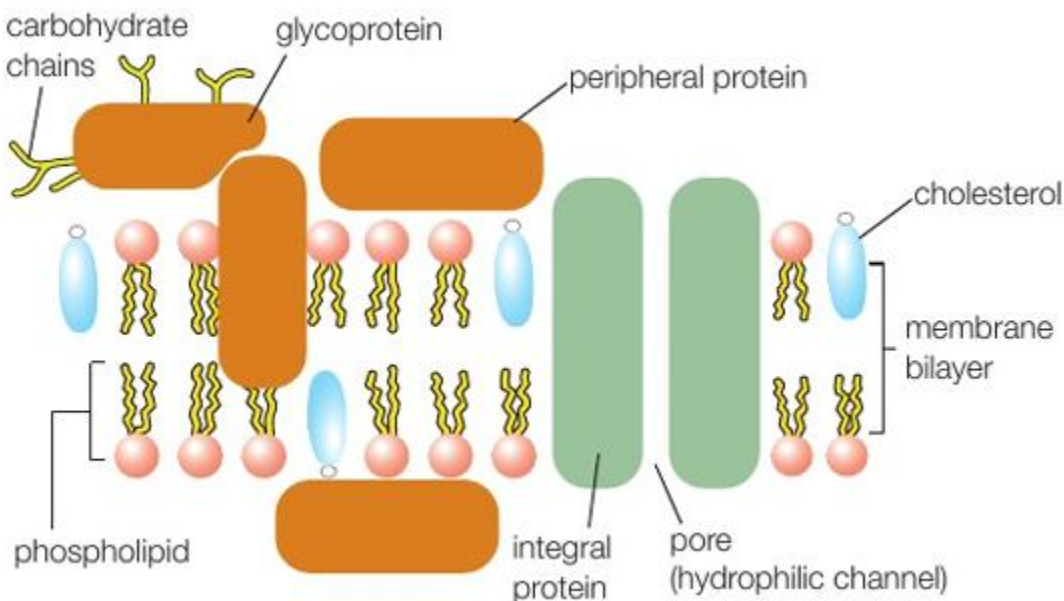
Implication:

- Increased surface area and gradient = faster diffusion
- Greater thickness = slower diffusion

2.2 Fluid Mosaic Model of Membrane Structure

Overview:

The cell membrane is described by the **fluid mosaic model** proposed by **Singer and Nicolson (1972)**.



Main Components:

1. Phospholipid Bilayer:

- **Hydrophilic heads** face outward (towards water).
- **Hydrophobic tails** face inward (away from water).
- Creates a **semi-permeable barrier**:
 - Allows small, non-polar molecules (e.g., O₂, CO₂)
 - Blocks large or charged molecules (e.g., ions, glucose)

2. Proteins:

- **Integral (intrinsic)** proteins: span the bilayer (e.g., channel or carrier proteins)
- **Peripheral (extrinsic)** proteins: attached to the surface
- Roles:
 - Transport (facilitated diffusion, active transport)
 - Cell signaling
 - Cell recognition (glycoproteins)

3. Cholesterol:

- Inserted between phospholipids
- **Stabilizes membrane** by reducing fluidity at high temperatures and preventing stiffness at low temperatures

4. Carbohydrates:

- Attached to proteins (glycoproteins) or lipids (glycolipids)
- Roles:
 - Cell-cell recognition
 - Receptors

Why 'Fluid Mosaic'?:

- **Fluid**: phospholipids and proteins can move laterally within the layer.
- **Mosaic**: patchwork of proteins embedded in or on the bilayer randomly.

2.3 Movement of Substances Across Membranes

Passive Transport (No Energy Required)

1. Simple Diffusion:

- Movement of **small non-polar molecules** (O_2 , CO_2) directly through the bilayer.
- Moves **down concentration gradient**.
- No carrier proteins or ATP required.

2. Facilitated Diffusion:

- For **large or charged molecules** (glucose, ions).
- Uses **channel proteins** (for ions) or **carrier proteins** (for polar molecules).
- Still **passive**, moves down concentration gradient.

Active Transport (Energy Required)

- Moves substances **against their concentration gradient** (low $\rightarrow$ high).
- Requires:
 - **Carrier proteins** that act as **pumps**
 - **ATP** for energy
- Example: Na^+/K^+ pump in neurons

Bulk Transport (Vesicle-Mediated)

1. Endocytosis:

- Large particles (e.g., proteins, bacteria) are engulfed by membrane forming a **vesicle**.
- Requires ATP.
- **Phagocytosis** = cell eating (solids)
- **Pinocytosis** = cell drinking (liquids)

2. Exocytosis:

- Vesicles inside the cell fuse with the membrane and **release contents outside**.
- Example: secretion of extracellular enzymes, hormones

Factors affecting membrane permeability

- ❖ **Temperature:** Increasing temperature increases kinetic energy of phospholipid molecules, so interactions between them are broken down, causing pores to develop within the membrane
- ❖ **Alcohol:** dissolves fatty acid molecules in the phospholipids, forming pores within the membrane
- ❖ More cholesterol reduces membrane fluidity and makes it more rigid, so their movement is restricted, which makes the membrane less porous

2.4 Osmosis and Water Potential

Osmosis Definition:

- The **net movement of water molecules** from a region of **higher water potential** to a region of **lower water potential** through a **partially permeable membrane**.

Water Potential (Ψ):

- Measure of the tendency of water to move.
- **Pure water** has $\Psi = 0$ (highest possible).
- **Solutes lower water potential**, so solutions have **negative values**.
 - More concentrated solution $\rightarrow$ more negative Ψ .

Types of Solutions (Relative to Cell Cytoplasm):

Solution Type	Water Movement	Effect on Animal Cell	Effect on Plant Cell
Hypertonic	Out of cell	Shrinks (crenates)	Plasmolysis (membrane pulls away)
Hypotonic	Into cell	Swells and may burst	Turgid (normal for plants)
Isotonic	No net movement	Stable	Flaccid (not turgid)

Osmosis in Cells:

- In **animal cells**, no cell wall: lysis occurs in hypotonic solution.
- In **plant cells**, cell wall resists bursting $\rightarrow$ develops **turgor pressure**
- Turgor pressure keeps plant stems upright so that sunlight absorption by chlorophyll in leaves is maximized for photosynthesis (glucose formation)

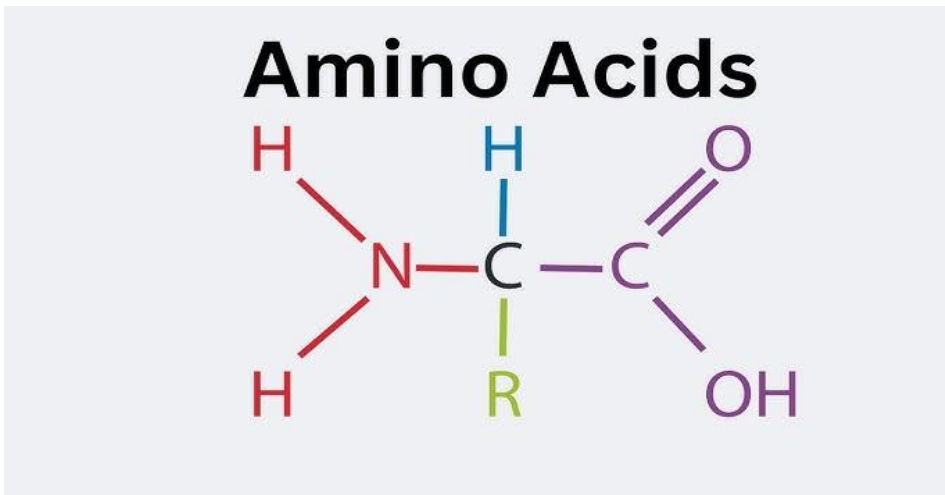
2.5 Structure and Types of Proteins

Proteins are made up of **amino acids** joined by **peptide bonds**, forming **polypeptides** that fold into specific 3D shapes. Their structure determines their function.

Amino Acid Structure:

Each amino acid has:

- A central carbon
- An **amino group** ($-\text{NH}_2$)
- A **carboxyl group** ($-\text{COOH}$)
- A **hydrogen atom**
- An **R group** (variable side chain)



2.6 Four Levels of Protein Structure

Level	Description	Bond/Interaction Involved	Example
Primary	Sequence of amino acids in a polypeptide chain	Peptide bonds	Insulin's chain sequence
Secondary	Folding into α-helix or β-pleated sheet	Hydrogen bonds between $\text{C}=\text{O}$ and $\text{N}-\text{H}$ groups	Keratin (α -helix), Silk (β -sheet)
Tertiary	Further folding into unique 3D structure by interactions between R-groups	Disulfide bridges, ionic bonds, hydrophobic interactions, hydrogen bonds	Enzymes, antibodies
Quaternary	Two or more polypeptide chains joined	Same as tertiary + interactions between subunits	Haemoglobin (4 chains + haem groups)

Key Notes:

- **Globular proteins** (e.g., enzymes, haemoglobin): soluble, functional.

- **Fibrous proteins** (e.g., collagen, keratin): insoluble, structural.

2.7 Properties and Functions of Globular and Fibrous Proteins

Globular Proteins:

- **Compact and spherical**
- **Water-soluble** (hydrophilic R-groups on outside)
- **Functional proteins:**
 - Enzymes: catalyse reactions
 - Antibodies: immune defense
 - Hormones: insulin, etc.
 - Transport: haemoglobin

Example: Haemoglobin

- **Quaternary structure:** 4 subunits, each with a **haem group**
- Each haem binds **1 O₂ molecule** → 4 O₂ per Hb

Fibrous Proteins:

- **Long, narrow, repetitive structure**
- **Insoluble** in water
- Provide **mechanical strength and support**
- **High tensile strength**

Example: Collagen

- Found in skin, tendons, ligaments
- Triple helix of polypeptide chains
- **High tensile strength**
- Every 3rd amino acid = **glycine** (smallest side chain) for tight coiling
- Cross-linking increases stability, large number of hydrogen bonds hold the polypeptide chains together in the helix
- Highly repetitive primary sequence with most common being hydroxyproline and proline
- Little to no tertiary structure

Feature	Haemoglobin (Hb)	Collagen
Type of Protein	Globular	Fibrous
Function	Transport of oxygen in the blood	Structural support in connective tissues (tendons, ligaments, skin)
Solubility	Soluble in water (hydrophilic R groups face outward)	Insoluble in water (due to tightly packed, hydrophobic nature)
Shape	Compact, spherical	Long, rope-like triple helix
Quaternary Structure	4 polypeptide chains (2 alpha, 2 beta), each with a haem group	3 polypeptide chains intertwined into a triple helix
Tertiary Structure	Well-defined, complex folding of each polypeptide	Little to no tertiary structure (each chain is elongated and remains extended)
Prosthetic Group	Yes – contains a haem group with Fe ²⁺ for O ₂ binding	No prosthetic group
Amino Acid Composition	Variable, allows precise folding	Highly repetitive sequence: Glycine every third residue , often with proline and hydroxyproline
Bonding	- Hydrogen bonds - Ionic bonds - Disulfide bridges - Hydrophobic interactions	- Hydrogen bonds (within helix) - Covalent cross-links between molecules (increases tensile strength)
Biological Role	Reversibly binds oxygen in lungs and delivers it to tissues	Provides tensile strength and structural integrity to body tissues
Example Location	Red blood cells	Skin, tendons, ligaments, cartilage

2.8 Enzymes: Structure and Function

Enzyme Basics:

- **Biological catalysts:** speed up reactions by **lowering activation energy**
- Do not get used up
- Highly **specific** due to their **active site shape**, so only one enzyme catalyses a single reaction

Mechanism of Action:

1. Lock-and-Key Hypothesis:

- Substrate fits perfectly into **complementary active site**.
- Forms **enzyme-substrate complex**
- Bonds in substrate are broken or formed → product released.

2. Induced Fit Hypothesis (more accurate):

- Active site **molds around substrate** slightly after binding.

- Enhances specificity and binding strength.

Enzymes are affected by:

- **Temperature**
- **pH**
- **Substrate concentration**
- **Enzyme concentration**
- **Presence of inhibitors**

Key Structural Requirements:

- Active site formed by **tertiary structure**.
- Any change in **primary structure** (e.g., mutation) can affect folding which changes the R groups which leads to different interactions opposed to original which results in a changed active site shape

2.9 Factors Affecting Enzyme Activity

1. Temperature

- **Rate increases** with temperature due to **greater kinetic energy** → more frequent enzyme-substrate collisions.
- **Optimum temperature**: varies by enzyme (e.g., human enzymes $\approx 37^{\circ}\text{C}$).
- **Above optimum**: H-bonds and ionic bonds break → enzyme **denatures** → active site loses shape (irreversible).

2. pH

- Each enzyme has an **optimum pH** (e.g., pepsin $\approx \text{pH } 2$; amylase $\approx \text{pH } 7$).
- Extreme pH changes **alter charges on amino acids**, disrupting **ionic bonds** and **active site shape**.
- Denaturation may occur if conditions persist.

3. Substrate Concentration

- **Increased substrate concentration** increases rate, up to a point.
- At **saturation**, all active sites are occupied — **maximum rate (V_{max})** is reached.

4. Enzyme Concentration

- Increasing enzyme concentration increases rate **if substrate is not limiting**.

- If substrate is limited, rate **plateaus**.

5. Competitive Inhibitors

Mechanism:

- **Inhibitor structurally resembles the substrate.**
- It **binds to the enzyme's active site, competing** with the substrate.
- Binding is **reversible** — inhibitor does **not permanently damage the enzyme**.

Effect on Enzyme Activity:

- **Fewer active sites** available for substrate binding.
- At **low substrate concentrations**, rate of reaction is reduced.
- At **high substrate concentrations, inhibitor effect is overcome**:
 - The substrate **outcompetes** the inhibitor.
 - **Maximum rate** can still be reached.

6. Non-Competitive Inhibitors

Mechanism:

- Inhibitor binds to an **allosteric site** (not the active site).
- **Does not resemble the substrate.**
- Binding **alters the tertiary structure** of the enzyme → **active site changes shape** → substrate cannot bind.
- Effect can not be reversed as active site is denatured.

2.10 DNA and RNA: Structure and Differences

Nucleotides:

Each nucleotide contains:

- A **phosphate group**
- A **pentose sugar**:
 - **Deoxyribose** in DNA
 - **Ribose** in RNA
- A **nitrogenous base**:
 - DNA: A, T, C, G

- RNA: A, U, C, G

DNA Structure:

- **Double-stranded helix**, antiparallel strands
- Sugar-phosphate backbone
- Bases pair via **hydrogen bonds**:
 - A-T (2 H bonds)
 - C-G (3 H bonds)
- Base pairing is **complementary**

RNA Structure:

- **Single-stranded**
- Shorter than DNA
- Ribose sugar instead of deoxyribose
- Base **uracil (U)** replaces thymine
- Not helical

Key Differences:

Feature	DNA	RNA
Sugar	Deoxyribose	Ribose
Bases	A, T, C, G	A, U, C, G
Strands	Double	Single
Function	Genetic information storage	Protein synthesis (mRNA, tRNA, rRNA)
Location	Nucleus (mainly)	Made in nucleus, functions in cytoplasm

2.11 DNA Replication

Occurs during **S-phase** of the cell cycle — essential before mitosis or meiosis.

Semi-conservative Replication:

Each daughter DNA contains **one original strand and one new strand**.

Process:

1. **DNA helicase**:
 - Breaks **hydrogen bonds** between base pairs.

- **Unwinds double helix**, creating **replication forks**.

2. **DNA polymerase:**

- Joins **DNA nucleotides** (from nucleoplasm) to **complementary bases** on the exposed strands.
- Joins adjacent nucleotides via **phosphodiester bonds**.
- Can only build from **5' to 3' direction**.
- Corrects mistakes and mutations in DNA synthesis

3. **Result:** Two identical DNA molecules.

2.12 Evidence for Semi-Conservative Replication – Meselson-Stahl Experiment

Method:

1. Bacteria grown in **15N** medium (heavy isotope) → all DNA heavy.
2. Moved to **14N** medium → allowed to replicate.
3. DNA extracted after each generation and separated via **centrifugation** in a density gradient.

Results:

- **Generation 1:** All DNA had **intermediate density** → **not conservative**.
- **Generation 2:** Two bands — one **light**, one **intermediate** → supports **semi-conservative replication**.

2.13 Genes, the Genetic Code, and Codons

Gene:

- A **sequence of DNA** that codes for the **amino acid sequence** of a specific **polypeptide** or **functional RNA** (e.g., tRNA, rRNA).

Genetic Code:

- **Triplet code:** Each amino acid is encoded by a group of **three bases** (a **codon**).
- The code is:
 - **Universal:** same codon = same amino acid in almost all organisms.
 - **Non-overlapping:** codons are read one at a time, not reused.

- **Degenerate:** most amino acids are coded by **more than one codon** — helps reduce the impact of mutations.
 - Example: Both GGU and GGC code for glycine.

Codons and Anticodons:

- **Codon:** A triplet of mRNA bases (e.g., AUG).
- **Anticodon:** Complementary triplet on **tRNA** (e.g., UAC for methionine).

2.14 Transcription and Translation (Protein Synthesis)

Overview:

Involves two main stages:

1. **Transcription** – nucleus
2. **Translation** – cytoplasm (at ribosome)

1. Transcription (in the nucleus):

1. **RNA polymerase** binds to **promoter region** of DNA.
2. DNA **unzips** — H-bonds between bases break.
3. **Template strand** (antisense strand) is read by RNA polymerase.
4. **Free RNA nucleotides** align with complementary bases.
 - A → U (uracil)
 - T → A
 - G ↔ C
5. RNA polymerase catalyzes the formation of **phosphodiester bonds** and checks for errors on transcribed mRNA
6. **mRNA** strand is formed and **detaches**, then **leaves the nucleus** via a nuclear pore.

2. Translation (in cytoplasm at ribosome):

1. **Ribosome** binds to mRNA at **start codon (AUG)**.
2. A **tRNA molecule** with complementary anticodon (UAC) brings **methionine**.
3. Another tRNA brings the next amino acid.

4. **Peptide bond** forms between amino acids (**enzyme + ATP required**).
5. Ribosome moves along mRNA, repeating the process.
6. The ribosome holds the mRNA in place until a peptide bond forms between the two amino acids, while the tRNA holds the amino acids in place until a peptide bond is formed
7. Translation continues until a **stop codon** (UAA, UAG, or UGA) is reached.
8. Polypeptide is released and folds into its functional structure.

Mutations and Their Effects on Protein Structure and Function

Mutation:

A change in the nucleotide sequence of DNA.

Types of Mutations:

Type	Description	Potential Effect
Substitution	One base replaces another	May change one amino acid (missense), or none (silent), or create a stop codon (nonsense)
Insertion	One or more extra bases added	Causes frameshift — changes every amino acid downstream Stop codon may appear, shortening the protein or preventing synthesis entirely Frameshift causes protein to be non functional
Deletion	One or more bases removed	Also causes frameshift unless in multiples of 3 Similar effects as insertion

1. Substitution Mutation

A single nucleotide is **replaced** by another. This affects **only one codon** and may have one of the following effects:

(i) Silent Mutation:

- New codon still codes for the **same amino acid** (due to the genetic code being **degenerate**).
- No effect on protein structure or function.

Example: AAA → AAG both code for **lysine**

(ii) Missense Mutation:

- Codon changes → **different amino acid** inserted.

- May cause **slight change in function**, **loss of function**, or **no functional change** (depends on importance of that amino acid in folding/active site).

Example: AAA → ACA changes **lysine** to **threonine**

(iii) Nonsense Mutation:

- Codon changes to a **stop codon** (e.g., UAG, UGA, UAA).
- Results in a **truncated protein**, often **non-functional**.

Example: UAC (tyrosine) → UAG (stop codon)

2. Insertion Mutation (Addition of a Base)

- Adds **one or more nucleotides** to the sequence.
- Causes a **frameshift** mutation.
- Shifts the **reading frame** one place forwards
- Results in **completely different amino acid sequence** and often an early stop codon.
- Protein is usually **non-functional**.

Example: Adding C between bases shifts triplet groups → different amino acids coded

3. Deletion Mutation (Loss of a Base)

- Removes one or more bases from the sequence.
- Causes a **frameshift**
- Shifts the **reading frame** one place backwards.
- Often results in a **non-functional protein** or premature stop codon.

Example: Deleting base 4 shifts all codons afterward → alters downstream amino acids

For both **deletion** and **insertion**, the closer the **mutation** to the **start** of the sequence, the **greater** the **effect**. This is because, both **deletion** and **insertion** affects all the bases after the mutation has taken place.

Effects on Protein:

- Change in **primary structure** so **different** amino acids → may affect **folding** → different **amino acids** have **different R groups** so different **interactions** which changes overall **tertiary structure** → changes **active site** (for enzymes) or binding site (for receptors).
- May cause:
 - **Loss of function** (e.g., non-functional enzyme)

- **Gain of function** (e.g., overactive protein in cancer)
- **Neutral** if mutation is silent or conservative substitution
- Protein is not **synthesised** at all in some cases of a stop codon being placed, or the protein is too short to function

2.15 Genetic Inheritance: Dominant, Recessive, Monohybrid Crosses

Key Terms:

- **Allele**: different forms of the same gene.
- **Homozygous**: both alleles are the same (AA or aa).
- **Heterozygous**: alleles are different (Aa).
- **Genotype**: genetic makeup (e.g., Aa).
- **Phenotype**: observable trait (e.g., attached earlobe).
- **Dominant allele**: expressed even if only one copy is present.
- **Recessive allele**: expressed only if both alleles are recessive.

Monohybrid Cross:

- Involves one gene with two alleles.
- Use **Punnett squares** to predict genotypic and phenotypic ratios.

Example: Tall (T) is dominant over short (t)

Parent genotypes: Tt × Tt		
	T	t
T	TT	Tt
t	Tt	tt

- Genotype ratio: 1 TT : 2 Tt : 1 tt
- Phenotype ratio: 3 Tall : 1 Short

2.16 – How a Gene Mutation in Cystic Fibrosis Affects Organ Systems

Gene Mutation:

- **CFTR gene** (on chromosome 7) codes for a **chloride ion channel** in epithelial cell membranes.
- Most common mutation = **ΔF508**: deletion of three bases → **loss of phenylalanine** at position 508.
- This leads to:
 - **Misfolding** of the protein
 - CFTR is **degraded by the cell** and **fails to reach membrane**

Effect on Protein Expression and Organ Function:

1. Gaseous Exchange System (Lungs)

- **Normal**: CFTR transports Cl^- ions out of epithelial cells into mucus → Na^+ follows → water follows by osmosis → mucus stays thin.
- **In CF**:
 - Cl^- transport fails → Na^+ and cytoplasm in cells become more highly concentrated than mucus
 - water enters cells from mucus by osmosis
 - Mucus becomes **thick and sticky**
 - **Cilia cannot move mucus effectively**

Consequences:

- Mucus accumulates in airways → **blocks bronchioles**
- Reduced gas exchange → **shortness of breath, hypoxia**
- **Traps bacteria** → recurrent lung infections (*Pseudomonas*)
- Chronic inflammation → **scarring (fibrosis)** and **bronchiectasis**

2. Digestive System

- **Pancreatic ducts** are blocked by mucus.
- Digestive enzymes **cannot reach duodenum** → **enzyme deficiency in gut**.
- Proteins, fats, and carbs are not digested efficiently.

Consequences:

- **Malabsorption**, poor growth, low energy
- **Steatorrhea**: fatty, foul-smelling stools
- Risk of **pancreatitis** and **diabetes** from pancreatic damage

3. Reproductive System

- **Males:**
 - **Vas deferens** is often absent or blocked → sperm cannot leave testes
 - Result: **infertility**
- **Females:**
 - **Cervical mucus** is abnormally thick → **impairs sperm movement**
 - Reduced chance of natural fertilization

2.17(i) – Genetic Screening: Uses

Genetic screening = testing DNA to detect specific **mutations or alleles** associated with inherited disorders (like CF).

1. Carrier Testing

- Identifies **asymptomatic individuals** who carry **one faulty CFTR allele**.
- Important for:
 - Family planning
 - Screening relatives of affected individuals
 - Allows to decide whether or not take child
 - Some people undergo this screening before marriage or planning a child to see whether their offspring is likely to have any genetic diseases

2. Pre-Implantation Genetic Diagnosis (PGD)

- Used during **IVF**.
- After fertilisation, embryos are grown to **8-cell stage**.
- A single cell is removed → DNA tested for CF mutations.
- Only **unaffected embryos are implanted**.

Benefits:

- Avoids passing on CF
- No risk of miscarriage (compared to invasive prenatal tests)

Limitations:

- Ethical concerns about embryo selection
- Expensive and emotionally demanding
- Embryo destruction may be considered as killing
- Unwanted information may be revealed which cause disputes (paternity etc)

3. Prenatal Testing**A. Amniocentesis:**

- Done at **15–20 weeks** of pregnancy.
- A needle withdraws **amniotic fluid** containing fetal cells.
- Risk: ~1% **miscarriage**

B. Chorionic Villus Sampling (CVS):

- Done earlier (around **10–13 weeks**).
- Takes a sample of **placental tissue** (chorionic villi).
- Slightly **higher miscarriage risk** (~1–2%) than amniocentesis.

Used to:

- Help parents make informed decisions
- Prepare to have child with genetic diseases
- Make treatment plan before birth

2.17(ii) – Implications of Prenatal Genetic Screening**Biological Implications:**

- Enables **early detection** of serious genetic conditions.
- Gives parents time to prepare medically, emotionally, and financially.

Risks:

- Both amniocentesis and CVS have **miscarriage risks**.
- Can result in **false positives** or **false negatives**.
- May lead to **termination decisions**, which can be distressing and controversial.

- Painful for mother if needle injection is involved

Psychological Implications:

- Anxiety during waiting period for results
- Emotional impact of a positive result
- Long-term guilt if decisions lead to termination

2.18 – Ethical and Social Issues in Genetic Screening

Ethical Issues:

Viewpoint	Key Considerations
Pro-life	Screening could lead to selective abortion, which some believe is morally wrong
Utilitarian	Screening prevents suffering and reduces medical burden — may be seen as ethical
Religious	Some faiths oppose interference in natural reproduction or embryo selection
Parental autonomy	Parents have the right to make informed choices about their child's health

Social Issues:

- **Discrimination:** Could lead to stigmatization of people with genetic conditions.
- **Pressure to screen:** Societal norms may pressure couples to undergo testing or terminate affected pregnancies.
- **Privacy and consent:** Genetic data may be misused (e.g., by insurers or employers).
- **Designer babies concern:** Fear that PGD could be extended beyond disease prevention to select non-medical traits.

Additional Technologies: Invasive vs Non-Invasive Screening Methods

Non-Invasive Techniques

- **Carrier Testing** – blood/saliva test (parents)
- **Ultrasound Scanning**
- **NIPT (Non-Invasive Prenatal Testing)** – cell-free fetal DNA in maternal blood
- **PGD (Pre-Implantation Genetic Diagnosis)** – during IVF, not invasive to mother

Invasive Techniques

- **Amniocentesis** – extraction of amniotic fluid

- **Chorionic Villus Sampling (CVS)** – placental tissue sample
- **Fetal Blood Sampling (Cordocentesis)** – blood from umbilical cord (*rare and high-risk*)

Comparison of Fetal Screening Technologies

Method	Type	When Done	Sample	Benefits	Limitations / Risks
Carrier Testing	Non-Invasive	Before conception	Blood or saliva	No risk, identifies risk in parents	Doesn't test fetus directly
PGD (IVF)	Minimally Invasive	Before implantation	Embryo cell	Prevents affected embryo implantation	IVF needed, ethical concerns
Ultrasound	Non-Invasive	Throughout pregnancy	Imaging	Detects physical defects, safe	Can't detect genetic mutations
NIPT	Non-Invasive	~10 weeks onward	Maternal blood	No miscarriage risk, early detection	Screening only, false positives, confirm with invasive test
Amniocentesis	Invasive	15–20 weeks	Amniotic fluid	Accurate genetic diagnosis	1% miscarriage risk, discomfort
CVS	Invasive	10–13 weeks	Placental tissue	Earlier results than amniocentesis	1–2% miscarriage risk
Fetal Blood Sampling	Invasive	~18+ weeks (rare)	Fetal blood (cord)	Confirms disorders when other tests unclear	Highest miscarriage risk (~2–4%)

Note: There is a chance for all methods to be **false positive** or **false negative**.

Edexcel IAL Biology Unit 1: Key Definitions Table

Term	Definition
Monosaccharide	A single sugar unit; the simplest type of carbohydrate (e.g., glucose, fructose).
Disaccharide	A carbohydrate formed by two monosaccharides joined by a glycosidic bond (e.g., sucrose, maltose).
Polysaccharide	A complex carbohydrate formed by many monosaccharides linked by glycosidic bonds (e.g., starch, cellulose).
Glycosidic bond	A covalent bond that joins carbohydrate (sugar) molecules together, formed by a condensation reaction.